

Telecommunication Network Assignment

Application Chosen: File Transfer Application

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Selected Application

- File Transfer Application
- Used for transferring data files
- Requires reliable communication
- Examples: PDF files, images, videos, documents

QoS Requirement and KPI

- Reliability – High
- Latency – Low
- Throughput – High
- Energy – Low consumption

Packetization and Addressing

- Large file divided into packets
- Packet contains: header, source address, destination address, payload, checksum
- Addressing identifies sender and receiver
- Supports routing and correct packet delivery

Routing and Data Link Layer

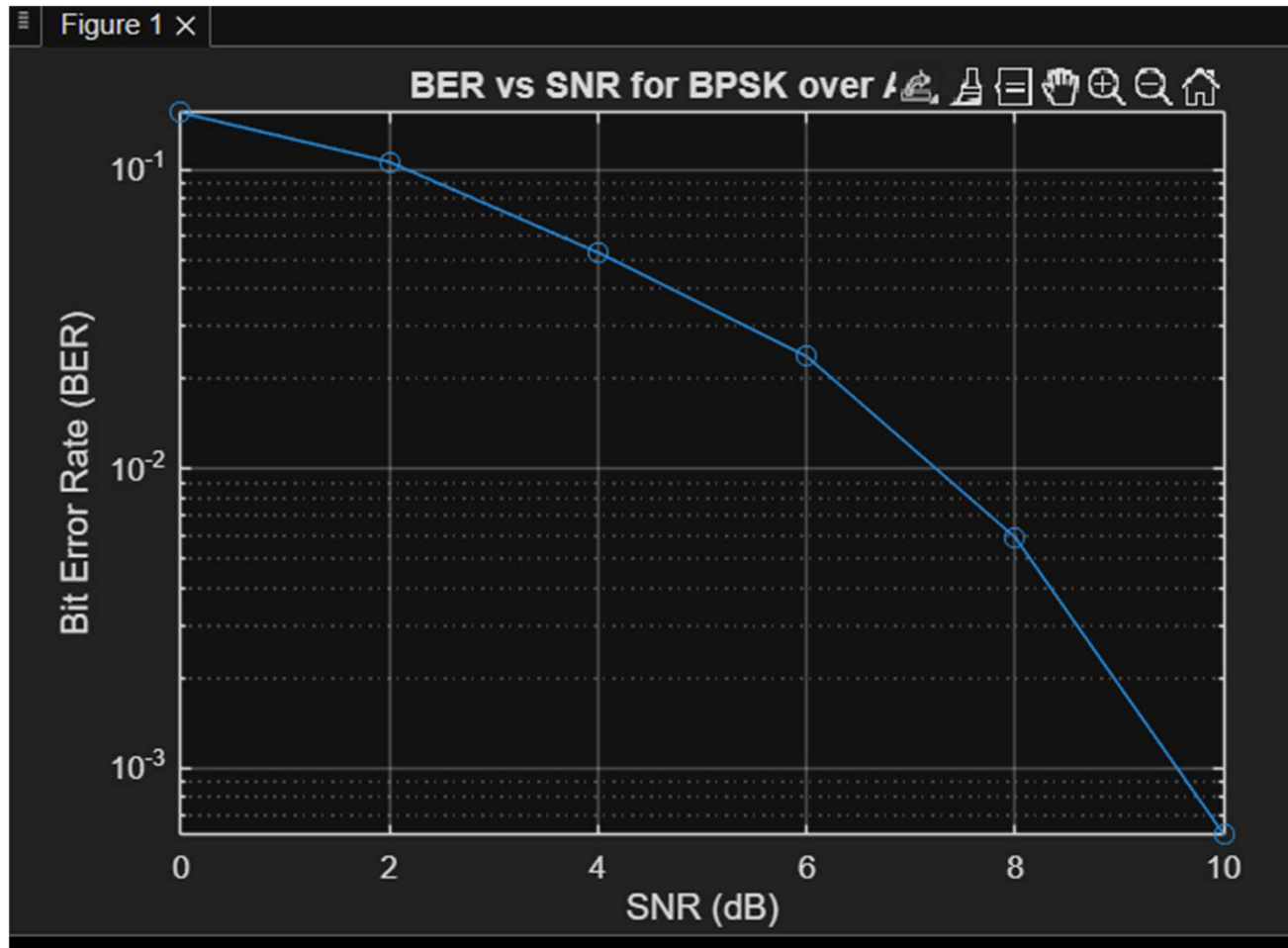
- Routing selects best path from sender to receiver
- Example: Sender → Router A → Router B → Receiver
- Selective Repeat ARQ used
- Only error frame retransmitted
- Improves reliability and throughput

Physical Layer and Performance Analysis

- Channel Used: AWGN Channel
- High SNR $\rightarrow$ Low BER
- Low SNR $\rightarrow$ High BER
- Expected Results: Low latency, high throughput, low BER, low packet loss

Result

- BER decreases as SNR increases
- High noise leads to more errors
- System performance improves at higher SNR



Conclusion

- Studied QoS, packetization, addressing, routing, ARQ, and AWGN channel
- Selective Repeat ARQ improves reliability and throughput
- Efficient routing reduces delay